

1. Operating System Overview

1.1 Definition, Components of OS, Types of Operating Systems

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What is an Operating System?

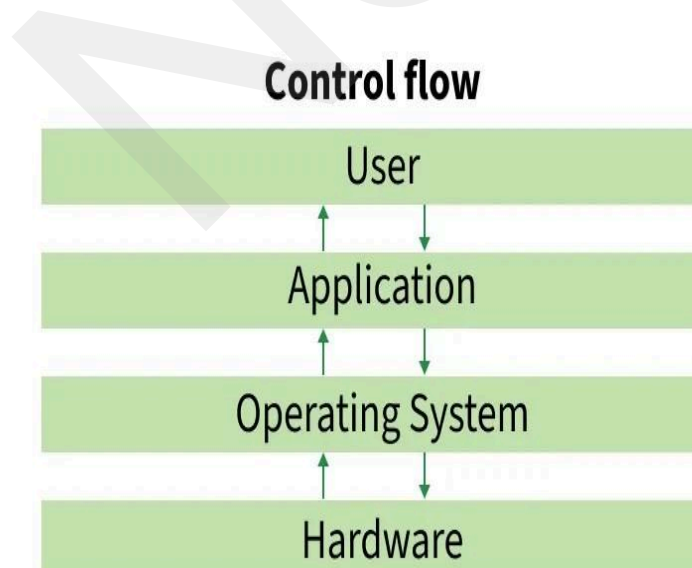
An Operating System (OS) is a software that acts as an interface between computer hardware components and the user.

Every computer system must have at least one operating system to run other programs.

Applications like Browsers, MS Office, Notepad Games, etc., need some environment to run and perform its tasks.

The OS helps you to communicate with the computer without knowing the computer's machine language.

It is not possible for the user to use any computer or mobile device without having an operating system.



An operating system (OS) is the program that, after being initially loaded into the computer by a boot program, manages all of the other application programs in a computer.

The application programs make use of the operating system by making requests for services through a defined application program interface (API).

In addition, users can interact directly with the operating system through a user interface, such as a command-line interface (CLI) or a graphical UI (GUI).

Types of Operating System (OS) :-

Following are the popular types of Operating System:

1. Batch Operating System
2. Multitasking/Time Sharing OS
3. Multiprocessing OS
4. Real Time OS
5. Distributed OS
6. Network OS
7. Mobile OS

1.Batch Operating System

Some computer processes are very lengthy and time-consuming.

To speed the same process, a job with a similar type of needs are batched together and run as a group.

The user of a batch operating system never directly interacts with the computer.

In this type of OS, every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.

2.Multi-Tasking/Time-sharing Operating systems

A time-sharing operating system enables people located at a different terminal(shell) to use a single computer system at the same time.

The processor time (CPU) which is shared among multiple users is termed as time sharing.

3.Real time OS

A real time operating system time interval to process and respond to inputs is very small.

Examples: Military Software Systems, Space Software Systems are the Real time OS example.

4.Distributed Operating System

Distributed systems **use many processors** located in different machines to provide very fast computation to its users.

5. Network Operating System

Network Operating System runs **on a server**. It provides the capability to serve to manage data, user, groups, security, application, and other networking functions.

6. Mobile OS

Mobile operating systems are those OS which are especially **that are designed to power smartphones, tablets,** and wearables devices.

Some most famous mobile operating systems are Android and iOS, but others include BlackBerry, Web, and watchOS.

Functions of Operating System

Below are the main functions of Operating System:



Functions of Operating System

In an operating system software performs each of the function:

1. Process management:- Process management helps OS to create and delete processes. It also provides mechanisms for synchronization and communication among processes.

2.Memory management:- Memory management module performs the task of allocation and de-allocation of memory space to programs in need of these resources.

3.File management:- It manages all the file-related activities such as organization storage, retrieval, naming, sharing, and protection of files.

4.Device Management: Device management keeps track of all devices. This module also responsible for this task is known as the I/O controller. It also performs the task of allocation and de-allocation of the devices.

5. I/O System Management: One of the main objects of any OS is to hide the peculiarities of hardware devices from the user.

6.Secondary-Storage Management: Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be stored in primary storage or cache so that a running program can reference it.

7. Security:- Security module protects the data and information of a computer system against malware threat and authorized access.

8.Command interpretation: This module is interpreting commands given by the and acting system resources to process those commands.

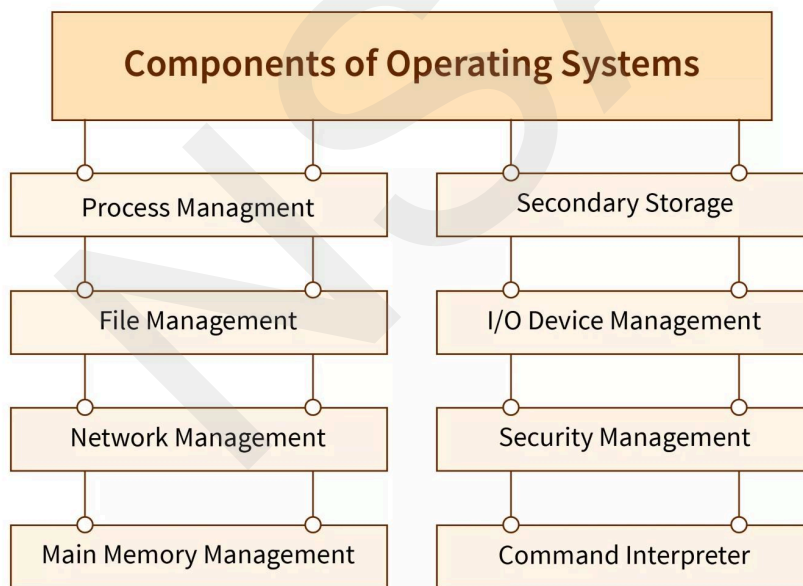
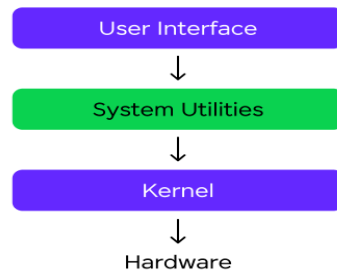
9.Networking: A distributed system is a group of processors which do not share memory, hardware devices, or a clock. The processors communicate with one another through the network.

10. Job accounting: Keeping track of time & resources used by various jobs and users.

11.Communication management: Coordination and assignment of compilers, interpreters, and other software resources of the various users of the computer systems.

Major Components of an Operating System :-

Major Components of an Operating System



Every operating system, no matter whether it's Windows, macOS, Linux, or Android, is built from several essential components that work together to make the system function efficiently.

These include the **Kernel**, **Process Management**, **Memory Management**, **File Management**, **Device Management**, **I/O System Management**, **Security and Protection System**, and **User Interface**.

Each component plays a specific role — from managing hardware and memory to ensuring data security and providing a user-friendly interface. Let's explore each one in detail to understand how they keep your system running smoothly.

1. Kernel

The kernel is the core part of an operating system — it's like the system's control center.

It manages the most critical operations such as allocating memory, scheduling tasks, and facilitating communication between software and hardware.

Without the kernel, the OS cannot interact with your computer's components.

Key Functions:

1. Controls CPU scheduling and task prioritization
2. Allocates and deallocates system memory
3. Facilitates interaction between applications and hardware
4. Handles system calls from programs

Example: When you open a music player, the kernel decides how much CPU and memory it should use while ensuring that other apps continue running smoothly.

2. Process Management

Every running program or application in your computer is called a process.

Process management handles how these processes are created, executed, and terminated, ensuring that multiple programs can run together without causing conflicts.

Key Functions:

1. Manages process creation, execution, and termination
2. Handles process synchronization and communication
3. Prevents deadlocks between competing tasks
4. Uses process scheduling algorithms to optimize performance

Example: When you're downloading a file while watching a video, process management divides CPU time so both tasks run efficiently.

3. Memory Management

Memory management takes care of how the system's main memory (RAM) is used.

It keeps track of every byte of memory and decides which process gets how much memory space.

Key Functions:

1. Allocates and releases memory as needed
2. Uses paging and segmentation to optimize performance
3. Manages virtual memory for large applications
4. Prevents processes from accessing each other's memory

Example: When you open several browser tabs, the OS assigns separate memory to each one so they don't interfere with one another.

4. File Management

All your documents, photos, apps, and system files are stored and organized by the file management system.

It maintains the structure of directories and ensures secure access to files.

Key Functions:

1. Organizes files into directories and subdirectories
2. Manages permissions and access control
3. Maintains metadata like file name, size, and creation date
4. Handles data retrieval and storage operations

Example: When you rename or move a file, the OS's file management system updates the directory and keeps your data organized.

5. Device Management

Device management ensures smooth coordination between the OS and all connected input/output devices — like keyboards, printers, USB drives, and monitors.

It uses device drivers to enable communication between hardware and software.

Key Functions:

1. Loads and manages device drivers
2. Handles interrupts from devices
3. Manages data transfer between hardware and memory
4. Tracks status and usage of connected devices

Example: When you connect a Bluetooth headset, the OS automatically detects it and sets it up using device management protocols.

6. I/O System Management

The Input/Output (I/O) system handles the flow of data between the computer's internal components and external devices.

It ensures that data transfers happen efficiently and reliably.

Key Functions:

1. Buffers and queues data for I/O operations
2. Synchronizes data transfer rates between devices
3. Ensures error-free transmission of data

Example: When you print a photo, the OS stores the file temporarily in a buffer before sending it to the printer when it's ready.

7. Security And Protection System

The security and protection system ensures that your computer remains safe from unauthorized access or harmful software.

It manages user permissions, detects threats, and encrypts sensitive information.

Key Functions:

1. Manages user authentication and access control
2. Encrypts files and confidential data
3. Monitors system activities for security threats
4. Prevents unauthorized system changes

Example: When your computer asks for admin permission before installing new software, the OS's security system is verifying user rights to prevent unsafe changes.

8. User Interface

The user interface (UI) is how users interact with the operating system.

It can be Graphical (GUI) or Command-Line (CLI) based, depending on the OS type.

Types of Interfaces:

Command-Line Interface (CLI): Text-based system used in Linux and UNIX

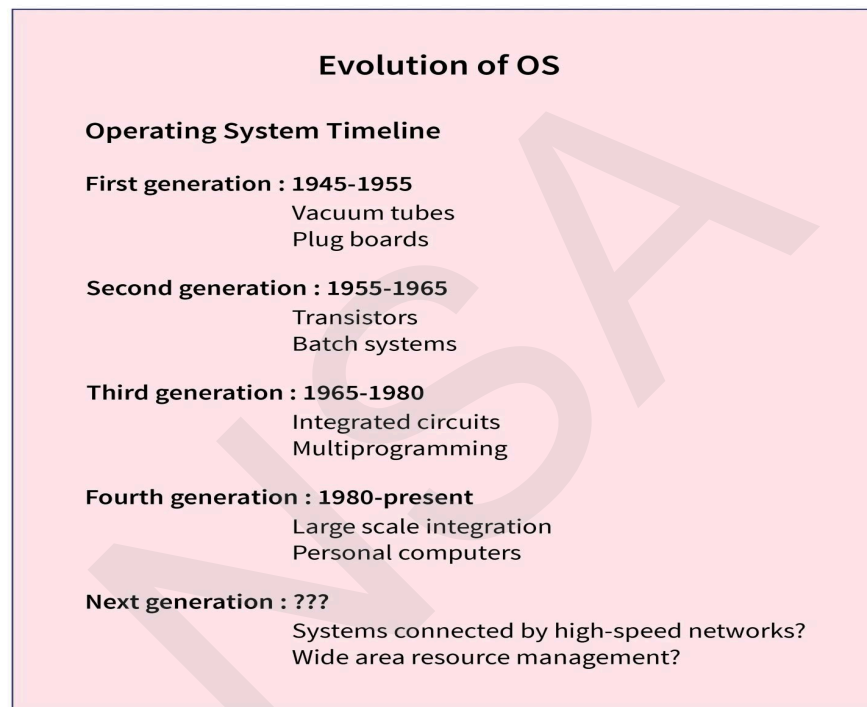
Graphical User Interface (GUI): Visual system with icons and windows, like in Windows or macOS

Example: When you click icons, drag files, or navigate folders, you're using the operating system's graphical interface to communicate with the system.

OS generation :-

OS generation refers to the historical evolution of operating systems, categorized into four main eras based on underlying hardware and features:

- First Gen (vacuum tubes, no OS),
- Second Gen (transistors, batch processing),
- Third Gen(integrated circuits, multiprogramming/time-sharing), and
- Fourth Gen(microprocessors, PCs, GUIs, networking, distributed systems).



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The Four Generations of Operating Systems

1. 1st Generation (1945-1955):

Hardware: Vacuum tubes, plugboards.

OS: No operating system; manual loading and execution, machine language programming.

Key Feature: Very slow, enormous machines, basic numerical calculations.

2. 2nd Generation (1955-1965):

Hardware: Transistors, magnetic tapes.

OS: Introduction of Batch Processing Systems (e.g., GM-NAA I/O) to automate job handling.

Key Feature: Improved efficiency, single job execution at a time, first commercial systems.

3. 3rd Generation (1965-1980):

Hardware: Integrated Circuits (ICs).

OS: Multiprogramming and Time-Sharing systems (e.g., Multics, early Unix).

Key Feature: Multiple jobs/users could share the CPU, faster processing, foundation for modern OS.

4. 4th Generation (1980-Present):

Hardware: Microprocessors, Large-Scale Integration (LSI).

OS: Personal Computers (PCs), Graphical User Interfaces (GUIs like Windows, macOS), Networking, Distributed Systems.

Key Feature: User-friendly, widespread adoption, mobility, internet connectivity (e.g., Linux, iOS, Android).

5. Beyond the Generations : Future

Modern OS development continues with focus on mobile (iOS, Android), cloud computing, virtualization, and AI integration, building upon the foundations laid by these generations.

Operating System as a Resource Manager :-

A resource in a computer system refers to any hardware or software component which are required to execute a process or program.

An Operating System (OS) acts as a resource manager by efficiently managing the computer's hardware (CPU, memory, I/O, storage) and software resources.

It tracks resource usage, allocates them as needed (request, use, release), schedules processes, manages memory (like virtual memory), and handles file systems, ensuring stability, security, and optimal performance.

The OS manages the following key resources:

Resource Type	Examples
Processor (CPU)	CPU cores, scheduling time
Memory	(RAM)Primary memory, virtual memory
Storage (Disk)	HDD, SSD, file system
I/O Devices	Keyboard, mouse, printer, network devices
Software Resources	System libraries, shared applications

Key Resources Managed by an OS:

- 1. Processor (CPU):** Schedules processes to share CPU time effectively (Process Management).
- 2. Memory (RAM):** Allocates memory to programs, manages virtual memory (using disk space), and protects memory from being accessed by other processes (Memory Management).
- 3. Input/Output Devices (I/O):** Manages access to printers, keyboards, disks, and network interfaces, handling drivers and requests.
- 4. Files & Storage:** Organizes data into files and directories, managing access and permissions (File System Management).
- 5. Network Resources:** Controls network access and communication.

How it Works:

Abstraction: Hides complex hardware details, presenting simplified interfaces (like files instead of disk sectors) for applications.

Allocation: Assigns resources (CPU time, memory blocks) to processes when requested.

Scheduling: Decides which process gets to use the CPU next.

Synchronization: Prevents multiple processes from interfering with each other or corrupting data.

Deallocation: Reclaims resources when a process finishes or releases them.

Monitoring & Protection: Tracks usage, detects errors, and enforces security to prevent misuse.

Why it's Important:

Efficiency : Maximizes hardware utilization.

Fairness : Ensures all applications get a fair share of resources.

Stability : Prevents crashes by isolating processes.

Usability : Makes complex hardware easy for users and programmers to handle.

The OS plays a critical role in managing resources through the following functions:

1. CPU (Processor) Management

The OS schedules processes and allocates CPU time efficiently.

CPU Scheduling Algorithms: FCFS, Round Robin, Priority Scheduling, Shortest Job Next (SJN).

Context Switching: Switches between processes to ensure multitasking.

Example: Running multiple applications like a browser and a video player.

2. Memory Management

The OS efficiently allocates and deallocates memory to processes.

Memory Allocation Strategies: Paging, Segmentation, Virtual Memory.

Prevents memory leaks and fragmentation.

Example: Running a large application by swapping parts of it into virtual memory.

3.Storage (File System) Management

The OS organizes and manages disk storage.

Manages file creation, deletion, and access permissions.

Implements file systems like NTFS, FAT32, ext4.

Example: Saving, retrieving, and organizing files on a hard drive.

4.Device (I/O) Management

The OS coordinates communication between the system and hardware devices.

Manages device drivers for peripherals (e.g., printers, scanners).

Handles I/O operations using buffering, spooling, and caching.

Example: Printing a document while using other applications.

5. Process Management

The OS controls and executes multiple processes efficiently.

Process Scheduling: Determines which process gets CPU time.

Inter-Process Communication (IPC): Enables processes to communicate.

Example: Running a word processor while listening to music.

6.Security & Access Control

The OS protects resources from unauthorized access.

User authentication and permissions.

Firewall and encryption for network security.

Example: Preventing unauthorized access to files.

3. Why is Resource Management Important?

Efficient resource management ensures:

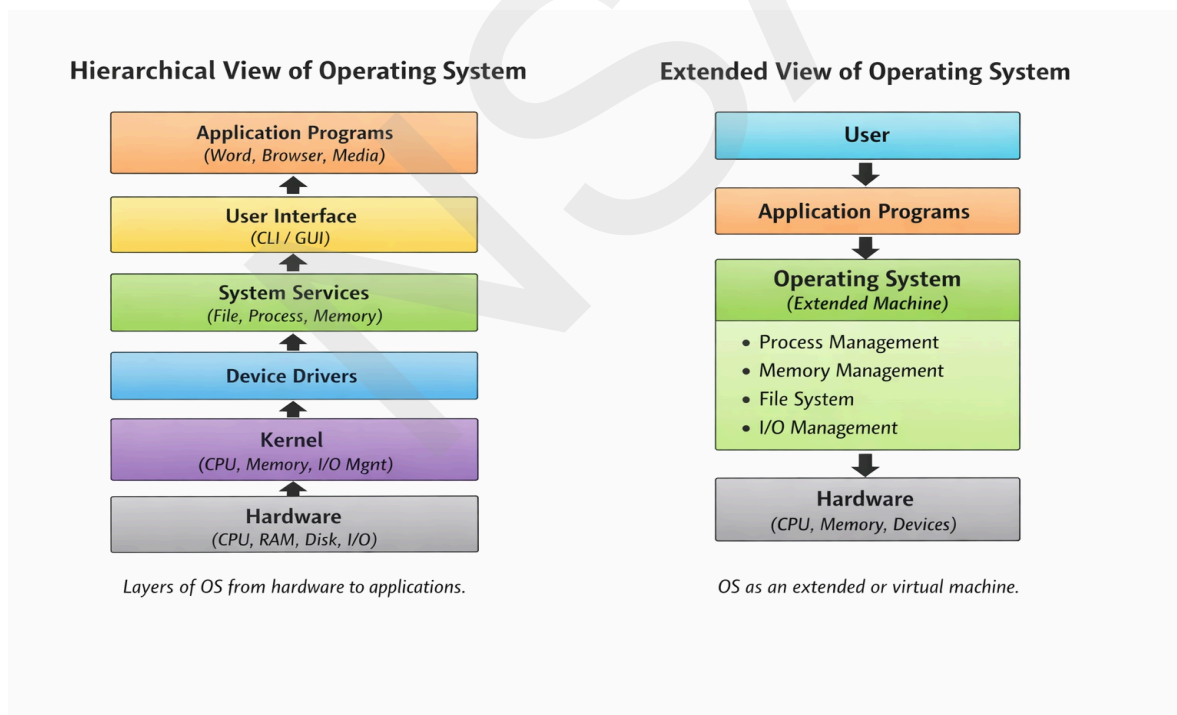
Optimized system performance – Prevents bottlenecks and overloading.

Fair resource allocation – Ensures all processes get necessary resources.

Prevention of deadlocks – Avoids processes from indefinitely waiting for resources.

Security and stability – Protects data and ensures smooth execution of processes.

Operating system– Hierarchical And Extended machine view :-



1. Hierarchical View of Operating System

In the hierarchical view, the Operating System is seen as a layered structure.

Each layer performs a specific function and interacts only with the layer directly above or below it.

Structure (Bottom to Top)

1.Hardware :- CPU, memory, I/O devices

2.Kernel :- Core part of OS

Manages CPU scheduling, memory management, and device control

3.Device Drivers :- Control and communicate with hardware devices

4.System Services:- File system, process management, security

5.User Interface :- Command Line Interface (CLI) or Graphical User Interface (GUI)

6.Application Programs :- Word processor, browser, media player, etc.

Advantages

1. Easy to design and debug
2. Clear separation of functions
3. Better security and reliability

Disadvantages

1. Slower performance due to multiple layers
2. Difficult to define exact layer boundaries

2. Extended View of Operating System

In the extended view, the Operating System is considered as an extended or virtual machine that hides hardware complexity from the user.

Key Idea

- OS acts as an interface between user and hardware
- Provides a convenient environment to execute programs

Functions Provided by OS (Extended View)

1.Process Management : Creation, execution, and termination of processes

2.Memory Management : Allocation and deallocation of memory

3.File Management : Organizing, storing, and accessing files

4.I/O Management : Handling input and output devices

5.Security and Protection : User authentication and access control

Example

- User opens a file → OS manages disk access
- User runs a program → OS allocates CPU and memory

Advantages

- User does not need to understand hardware details
- Makes computer system easy to use

Difference Between Hierarchical and Extended View :-

Aspect	Hierarchical View	Extended View
Focus	Internal structure	User perspective
Representation	Layered architecture	Virtual machine
Complexity	Shows OS internals	Hides hardware details
Usage	System design	System operation

Operating-System Services :-

“Operating System Services are a set of functions provided by the OS to manage hardware, run applications, and enable user interaction with the system.”

Operating system services and system calls work together to allow programs to interact with hardware efficiently, providing essential functions like file access, process scheduling, and memory management.

System calls are the methods by which a program requests services from the operating system.

Example of System Call

Example: Saving a file

- You type text in an application.
- You click Save.
- The application uses a system call like write().
- The Operating System saves the data to the disk.

★ Here, write() is a system call.

Operating System (OS) services are essential functions that manage computer hardware and software, providing a user-friendly environment by handling tasks like running programs, managing files, controlling I/O, allocating resources (CPU, memory), ensuring security, detecting errors, and facilitating communication between processes.

These services act as an intermediary between the user/applications and the hardware, ensuring smooth, efficient, and secure system operation.

Operating system is software that acts as an intermediary between the user and computer hardware.

It is a program with the help of which we are able to run various applications. It is the one program that is running all the time.

Every computer must have an operating system to smoothly execute other programs.

- The OS coordinates the use of hardware and application programs for users.
- It provides a platform for running application software.
- It is a set of special programs that help the computer function properly.
- It controls input–output devices and system operations.
- It manages program execution and file handling.



OS Services

1. Program Execution

The Operating System manages program execution by loading programs into memory and scheduling them using algorithms like FCFS, SJF. It ensures smooth execution of user and system programs while preventing deadlocks where processes wait indefinitely. The OS also allocates system resources efficiently for proper functioning.

- Loads programs into memory for execution.
- Uses CPU scheduling algorithms (FCFS, SJF, etc.) to decide execution order.
- Handles and prevents deadlocks during execution.
- Manages system resources for efficient overall functioning.

Inside the computing unit, user program modules and system processes need to run smoothly to ensure optimal performance, and this is made possible by the program execution service. Whenever a user runs a program, the OS starts loading it from secondary storage into main memory.

By doing this, it establishes the execution environment and allocates the resources needed to maintain the program's sequential execution order and handle errors.

Functions

- To maintain system efficiency, this service manages crucial elements such as process creation, scheduling, and termination.
- It also maintains the state of the process and the execution order of the program, and prioritizes tasks based on their criticality.
- Identifies and assesses the errors and bugs to ensure consistent program execution.

Examples

- Opening and running Microsoft Word or Google Chrome on a computer.
- Executing a video game like GTA or Minecraft.

2. Input Output Operations

The Operating System manages all input-output operations and enables communication between users, programs, and device drivers.

Device drivers act as the software interface for hardware, allowing the OS to control devices smoothly.

The OS also provides programs with safe and timely access to input-output devices whenever required.

- Manages all input-output operations.
- Communicates with hardware through device drivers.
- Ensures proper coordination between devices and the system.
- Provides programs controlled access to input-output devices.

This service focuses on handling data flow within the computing system, i.e., how data is transmitted or routed between the main source and external devices such as keyboards, printers, monitors, and storage drives. In simple terms, it acts as middleware between the hardware and software parts.

But as we know, these external devices operate at different speeds, so to avoid delays or conflicts, the input/output (I/O) operations maintain the workflow timing and data-flow pace.

Functions

- This service buffers and queries data to align the workflow of external devices with the system's speed.
- It provides a consistent interface for accessing all the I/O devices.
- It actively tracks the device status and ensures smooth data exchange between the user and hardware.

Examples

Typing on a keyboard and seeing letters appear instantly on the screen.

Printing a document using a printer connected to your computer.

3. Communication Between Processes

The Operating System manages communication between processes and enables smooth data transfer among them. It handles both local communication on the same computer and communication across networked systems. This ensures efficient and coordinated exchange of information.

- Manages inter-process communication (IPC).
- Enables data transfer between processes.
- Supports communication across networked computers.
- Ensures secure and efficient information exchange.

Functions

- It provides a flexible pathway for both one-way and two-way communication, i.e., forwarding information in one direction and receiving responses from other sources.
- It also synchronizes different processes, allowing them to work together without any obstruction.
- It effectively manages notifications and signals between processes, ensuring a robust communication environment.

Examples

- Chat applications where users exchange messages in real time.
- A media player process interacts with the audio driver process to play sound.

4. File Management

The Operating System manages all file operations by granting programs the required access permissions such as read-only or read-write. It allows users to create, delete, store, and organize files across devices like, floppy disk or pen drives. The OS also decides how data should be stored and manipulated for efficient file management.

- Grants programs file access with permissions like read-only or read-write.
- Allows users to create, delete, and organize files.
- Manages file storage across devices such as hard disks and pen drives.
- Decides how data is stored and manipulated for efficient file management.

Functions

- It effectively tracks the exact locations of files on storage devices, enabling instant access and enhancing the user experience.
- During file manipulation and sharing, it ensures information consistency.
- To allow users to identify file formats, it also handles file naming conventions and extensions.
- It helps perform file manipulation tasks such as creating, saving, updating, reading, and deleting.

Examples

- Copying photos from a camera to your computer.
- The OS automatically manages and updates system log files.

5. Memory Management

The OS manages memory like a cricket team manager choosing players. Just as the manager decides whether a player enters the playing 11 or 15 based on performance, the OS checks whether a new program meets the requirements for memory. If it does, the OS decides how much memory the program needs and loads it into an appropriate location, preventing unnecessary memory usage.

- OS checks whether a new program meets the requirements to get memory.
- It decides how much memory the program needs.
- Loads the program into a suitable memory location.
- Prevents programs from using unnecessary or extra memory.

The memory management service in the operating system (OS) decides how the computing unit will use memory and how to allocate memory to various programs. After a program is executed, the OS allocates it a certain percentage of the total memory and actively monitors which parts of the memory are in use and which are free; accordingly, it deallocates the memory once the program completes.

Functions

- It automatically handles memory allocation and deallocation.
- When physical memory is full or excessive data is stored, it swaps data between RAM and storage.
- It also maintains a comprehensive record of which memory sections are reserved, consumed, or free.

Examples

- Running multiple apps like a browser, music player, and text editor at the same time.
- Using virtual memory to run heavy applications when RAM is limited.

6. Process Management

Process management can be understood by imagining the kitchen stove as the CPU and the chef as the Operating System. The chef must cook multiple dishes (programs) and ensure no single dish takes too much time so that every dish gets a fair chance to be cooked. In this way, the OS schedules and manages all programs efficiently to keep the system running smoothly.

- The CPU is like the kitchen stove where execution happens.
- OS (chef) ensures each program gets fair execution time.
- Prevents any program from occupying the CPU for too long.
- Schedules all programs efficiently to keep the system running smoothly.

Process management in the operating system (OS) allows multiple programs to run simultaneously without technical overlap or conflict. During this process, each active program module is allocated sufficient processing time and system resources to operate and communicate with other processes.

Functions

- As this service is responsible for executing processes efficiently, it helps detect and handle technical issues such as deadlocks, preventing unnecessary blocking of system resources.
- It stores the essential information for every process that executes, such as its ID, priority level, and current state.
- It can switch between processes quickly and efficiently.

Examples

Watching a YouTube video while downloading files in the background.

Switching between open applications such as a web browser, a Word document, and a calculator.

7. Security and Privacy

Security

The Operating System keeps the computer safe from unauthorized users by adding protective security layers. It defends the system from threats like viruses and hackers through tools such as firewalls and antivirus software. These measures ensure the safety of the computer and its data.

- Blocks unauthorized access to the system.
- Protects against viruses and malware.
- Uses firewalls and antivirus tools for defense.
- Ensures overall system and data safety.

As we know, in this software-driven world, the majority of services and tasks are performed online, creating a significant risk of cyberattacks.

To prevent data breaches or unauthorized access to your computer system, the operating system (OS) adds a security layer using authentication techniques such as passwords or biometric scans.

The security layer is integrated to protect files, memory, and devices from misuse.

Functions

- It safeguards system files and folders from accidental or intentional hindrances.
- It keeps track of all audit logs to monitor user activities and maintain accountability.
- The security service layer also uses encryption to protect sensitive user information and business data stored or transmitted to other systems.

Examples

- Logging into your computer using a password or fingerprint.
- Antivirus software blocks unauthorized access to important files.

Privacy

The Operating System helps keep personal information private, similar to locking a door that only you can open. It ensures that sensitive files and data remain hidden from others. In this way, the OS respects user confidentiality and protects important information.

- Keeps personal data hidden from others.

- Provides access control options like passwords.
- Ensures only authorized users can view files.
- Protects sensitive information from misuse.

8. Resource Management

The Operating System manages how system resources are shared among different processes. It allocates CPU time using CPU Scheduling Algorithms, handles memory management, and controls input-output devices. By deciding which process gets which resource, the OS ensures proper and efficient use of all available resources.

- Shares system resources among multiple processes.
- Allocates CPU time using scheduling algorithms.
- Manages memory and input-output devices.
- Ensures efficient and proper use of all resources.

When multiple users and programs are active at the same time in a system, each process will use many different resources, such as CPU, memory, and input/output devices. To ensure that these resources are shared efficiently among processes and users, the operating system (OS) implements scheduling algorithms that help each program module satisfy its requirements without locking other tasks or causing resource conflicts.

Functions

- It constantly monitors and tracks the resource usage to identify potential vulnerabilities.
- To enhance operational efficiency and minimize the system idle time, it optimizes the allocation process.
- It strongly supports the scheduling policies for determining the quality of resource sharing.

Examples

- The OS is assigning CPU time to a video editing software while other apps run in the background.
- Allocating memory space to a new browser tab while keeping others active.

9. User Interface

The Operating System provides a user interface that allows users to interact with the system either through a command-line interface (CLI) or a Graphical User Interface (GUI). In a CLI, the command interpreter executes user-typed commands. A GUI offers a mouse-based window and menu system that makes interaction easier and more visual.

- Provides both CLI and GUI for user interaction.
- CLI uses a command interpreter to run user commands.
- GUI offers windows, icons, and menus for easy use.
- Helps users communicate smoothly with the operating system.

The user interface (UI) service is one of the operating system services responsible for handling user activities and interactions with device screens. The UI service allows users to communicate with the OS through commands or triggered events, enabling them to easily perform tasks such as managing files, opening programs, and sending data without needing to understand the complex background processes.

So basically, there are two types of interfaces via which users can interact: Command-Line Interface (CLI) and Graphical User Interface (GUI).

CLI is for instructing the machine by typing text commands, and GUI is where users interact through visual icons, windows, and menus.

Functions

- Ensures a consistent and responsive experience across applications.
- Acts as a communication layer between the user and system functions.
- Accepts user input from devices such as keyboards, mice, or touchscreens.

Examples

- Typing commands in the Linux terminal to manage files.
- Navigating apps and settings through touch gestures on an Android phone.

10. Networking

This service enables communication between devices on a network by managing connections and controlling data exchange. It allows devices to connect to the internet, send and receive data packets, and maintain stable communication. Through proper network management, the system ensures smooth and reliable data transfer.

- Manages network connections between devices.
- Supports internet access and communication.
Sends and receives packets reliably.
- Ensures smooth and stable network operations.

11. Error Handling

The Operating System detects and handles errors occurring in the CPU, memory, or input-output devices to keep the system stable.

It prevents frequent errors, avoids deadlocks, and fixes issues that appear during program execution.

Error handling is an operating system service that detects, reports, and resolves errors that occur during program execution or system operations.

Errors can arise from hardware failures, software bugs, incorrect user commands, or unexpected conditions.

The OS continuously monitors the system to identify these problems early and take corrective actions to prevent crashes or data loss.

This service ensures that the system remains stable and reliable, and continues to function smoothly even when issues occur.

Functions

- Displays clear error messages or warnings to inform the user.
- Isolates damaged components or files to maintain system integrity.
- Helps software developers identify and fix issues through error codes or logs.

Examples

An application shows “Not Responding,” while the OS offers options to close it. The OS automatically recovers after an unexpected device disconnects.

12. Time Management

Imagine a traffic light as the Operating System, guiding cars (programs) on when to stop, wait, or move.

Red represents the simple queue, yellow is the ready queue, and green means the program is under execution.

By switching lights at proper intervals, the OS ensures all programs move smoothly without traffic or delays.

- Red light = programs waiting in the simple queue.
- Yellow light = programs in the ready queue
- Green light = programs under execution.
- OS changes states at proper intervals to ensure smooth program flow.